# Auditing the Costs of Multi-Group Fairness Repair

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Equal opportunity requires a classifier to have the same true-positive rate across demographic groups. When fairness concerns multiple demographic attributes, parity for each attribute separately does not ensure parity for their intersections, such as groups defined jointly by race and sex. We study postprocessing for intersectional equal opportunity, focusing on the accuracy loss and withdrawal of previously favorable predictions that accompany fairness repair. We audit three published approaches on Credit Default and COMPAS, measuring marginal and intersectional TPR gaps, predictive accuracy, and withdrawals on both repair and test data. We then introduce Retain-EG, which reweights the classifiers returned by an exponentiated-gradient reduction to reduce withdrawals under empirical fairness and accuracy constraints. The baseline methods achieve intersectional parity on the repair data, but substantial test disparities remain on COMPAS. Methods with smaller test disparities can also have lower accuracy and withdraw more favorable predictions. Retain-EG reduces test withdrawals in all twenty model–split fits with nearly unchanged average accuracy, although some subgroup outcomes worsen. These results show the need to assess both accuracy and withdrawal costs when evaluating intersectional fairness repair.

Additional Key Words and Phrases: equal opportunity, intersectional fairness, postprocessing, decision withdrawals

## 1 Introduction

Equal opportunity (EO) requires a classifier to have the same true-positive rate across demographic groups, so that individuals with a favorable true label have an equal chance of receiving a favorable prediction regardless of group membership. For a trained classifier, EO can be pursued through postprocessing, which adjusts decision thresholds or prediction probabilities across groups. When several demographic attributes matter, enforcing EO for each attribute separately may leave disparities among their intersections. Intersectional EO addresses this by applying the fairness constraints to groups defined jointly by these attributes, such as race and sex. However, satisfying these constraints does not by itself ensure that a repair avoids unnecessary accuracy loss or withdrawals of previously favorable predictions.

Consider eight applicants who all have positive labels, split into two groups of four. An initial classifier approves three in the first group and one in the second. One repair equalizes their TPRs at 50% by withdrawing one approval from the first group and granting one in the second. Another repair reaches the same rates by withdrawing two approvals from the first group, approving its previously rejected applicant, and granting one approval in the second group. Both repairs have the same TPRs and accuracy, but they withdraw 25% and 50% of the four original approvals, respectively. The second repair therefore withdraws an additional approval without improving either fairness or accuracy. This difference is invisible to the final confusion matrices. Appendix **??** illustrates the example.

We ask how existing methods for intersectional EO affect predictive accuracy and withdraw previously favorable predictions, and whether these costs can be reduced while meeting the fairness target. We evaluate accuracy and withdrawals as two distinct outcomes: withdrawing an incorrect favorable prediction improves accuracy, whereas withdrawing a correct one reduces it. The audit treats these outcomes as equally relevant costs of fairness repair. Our extension then tests one concrete intervention: reducing withdrawals while placing explicit limits on changes in fairness and accuracy.

We audit three methods on Credit Default and COMPAS using the same logistic-regression (LR) and histogram-gradient-boosting (HGB) scores: Hardt–Price–Srebro postprocessing [3], Morina et al.'s RTDP [5], and Agarwal et al.'s exponentiated-gradient (EG) reduction [1]. Our audit examines TPR gaps for individual sensitive attributes and their

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intersections, accuracy loss, and withdrawals of original positive predictions. We assess accuracy and withdrawals both overall and within groups, and compare all three dimensions on repair and test data. The results reveal trade-offs between fairness, accuracy, and withdrawals, with costs that are unevenly distributed across groups. They also show that intersectional parity achieved on the repair data can leave substantial disparities on unseen data.

Building on this audit, we introduce Retain-EG to reduce the costs of intersectional fairness repair. Retain-EG adjusts the weights of EG's existing classifiers to reduce withdrawals while limiting changes in fairness and accuracy on both repair and validation data. It reduces test withdrawals across all twenty model–split fits, with nearly unchanged average accuracy. However, improvements in test fairness vary, and some groups still experience accuracy losses. Our study combines an audit of the accuracy and withdrawal costs of existing EO methods with a targeted modification that reduces withdrawals while preserving average predictive performance.

## 2 Related Work

Hardt et al. [3] introduce postprocessing methods for equal opportunity and equalized odds. Their approach adjusts a trained predictor to meet group-level rate constraints. Morina et al. [5] address intersectional fairness using group-specific thresholds and randomized predictions. Agarwal et al. [1] formulate fair classification as a constrained optimization problem and reduce it to cost-sensitive learning, producing a mixture of classifiers. These methods provide the baselines for our audit. We compare their accuracy and withdrawal costs when applied to intersectional EO, using a common set of underlying prediction scores.

Fairness across individual demographic attributes does not necessarily protect their intersections. Kearns et al. [4] demonstrate this limitation and develop methods for protecting richer families of subgroups. We focus on intersections explicitly defined by the sensitive attributes in each dataset. This scope allows us to inspect the changes experienced by each included group, while also making sample support a practical concern: an intersection can have few positive examples even when its parent demographic categories are large.

A separate concern is whether fairness achieved on one sample generalizes to new data. Cotter et al. [2] study this problem using separate samples for model and constraint learning, motivating our use of validation data when modifying a fitted repair. Our evaluation distinguishes this sampling issue from variability due to randomized predictions. A policy can equalize expected TPRs on its fitting sample yet have unequal expected TPRs on new records; it can also produce unequal realized TPRs when binary decisions are drawn for the original records.

*Gap analysis.* Accuracy is a central objective in existing fairness methods, but it does not capture which original favorable predictions are withdrawn. Repairs with the same fairness and accuracy can still reverse different individuals' decisions. We therefore audit accuracy loss and withdrawals as two distinct costs of intersectional EO, examining their distribution across groups and comparing repair-set and test-set outcomes. Building on this audit, Retain-EG reweights EG's existing classifiers to reduce withdrawals under explicit fairness and accuracy constraints.

## 3 Field Landscape

The project connects algorithmic fairness to broader questions about measurement, representation, and accountability. First, the choice of a target label determines what the audit calls a correct decision. No recorded recidivism and no credit default are observable benchmark outcomes, but neither fully describes an individual's circumstances or the consequences of granting or denying an opportunity. Accuracy and EO are therefore properties relative to these labels. A reduction in a statistical disparity should not automatically be interpreted as a reduction in social harm.

Second, group definitions and sample sizes determine whose outcomes are visible. Reporting only an overall accuracy or a separate gap for each demographic attribute can hide costs concentrated at intersections. At the same time, estimates for small intersections are less stable. Restricting an evaluation to groups with adequate sample support makes the reported rates easier to interpret, but leaves a coverage limitation that must remain visible. This tension links fairness evaluation to data quality and representation, beyond the choice of optimization algorithm.

Third, changing a deployed decision rule raises questions about accountability to affected individuals. A withdrawal may correct a prediction error, deny a previously favorable decision to a correctly classified person, or impose an administrative burden. Our benchmark measures decision reversals without assuming a particular welfare effect. Keeping withdrawals separate from accuracy allows the audit to show both how prediction quality changes and how those changes affect people who originally received favorable predictions. Together, these perspectives motivate an evaluation that reports the fairness target, the costs of meeting it, and the limits of what the data can establish.

## 4 Methods

### 4.1 Data and experimental design

We use UCI Credit Default [7] and COMPAS [6]. Credit contains 30,000 records; the favorable label is no default next month, and the sensitive attributes are sex and age below 40 versus at least 40. The base models exclude ID, sex, age, and the target column. COMPAS contains 6,172 records after filtering, with no recorded two-year recidivism as the favorable label. We audit race, sex, and their intersections, using prior-count, juvenile-count, and charge-degree features. These are our fitted models' predictions, not the proprietary COMPAS score. Favorable predictions represent hypothetical decisions.

For each dataset, five random seeds create disjoint training, calibration, repair, validation, and test partitions in proportions 50/10/15/10/15. We train LR and HGB, apply sigmoid calibration, then freeze the scores. The original classifier's threshold maximizes repair-set accuracy. All methods share the same scores, original predictions, and partitions, yielding twenty paired model–split fits. Repair data fit the fairness methods, validation data constrain Retain-EG, and test data evaluate final outcomes. We report descriptive paired comparisons because records recur across splits and models share partitions.

Headline intersections must have at least twenty repair-set positives. This retains four Credit intersections and six COMPAS intersections: African-American and Caucasian women and men, Hispanic men, and Other men. Marginal comparisons cover each dataset's sensitive attributes separately. An earlier sequential-repair experiment motivated comparing joint intersectional policies, because repairing one attribute could undo parity for another. Appendix **??** gives partition sizes, filtering and implementation details. Code and experiment artifacts: **[ADD GITHUB REPOSITORY URL]**.

### 4.2 Evaluation metrics

We evaluate three outcomes on both repair and test data. The *TPR gap* is the largest minus the smallest group TPR, computed separately for each sensitive attribute and for the supported intersections. The *accuracy drop* is the original classifier's accuracy minus the repaired classifier's accuracy on the same sample and group. We examine overall and within-group changes; the figure also reports absolute overall and worst-group accuracy. Worst-group accuracy is not the largest within-group accuracy drop, since the groups may start at different accuracy levels.

The *withdrawal rate* measures how often original favorable predictions are reversed. Let $b_i$ be the original binary prediction, $q_i$ the final probability of a favorable prediction, and $D_g$ the records in group $g$. Then

$$\widehat{W}_{g,D}(q;b) = \frac{\sum_{i\in D_g} b_i(1-q_i)}{\sum_{i\in D_g} b_i}. \tag{1}$$

The denominator includes both correct and incorrect original favorable predictions, so withdrawal differs from the false-negative rate. We report overall, within-group, and maximum-group withdrawal rates. Worst-group accuracy and maximum-group withdrawal are computed within each fit before averaging. All metrics use prediction probabilities, averaging over each method's randomization. Thus repair-to-test differences reflect sampling and policy generalization, not simulated binary draws. Appendix **??** gives the full definitions.

### 4.3 Baseline methods

Hardt's postprocessor chooses randomized score thresholds to maximize accuracy under TPR-parity constraints, treating intersections as sensitive categories. RTDP first selects an accuracy-maximizing threshold for each intersection, then optimizes approval probabilities above and below that threshold under equal-TPR constraints. EG reduces fair classification to cost-sensitive learning and returns a mixture of classifiers. Our oracle searches group-specific thresholds over frozen scores, making this application postprocessing. The methods use different policy classes and coverage rules; the audit compares their attained outcomes rather than solutions from an identical feasible set.

### 4.4 Retain-EG

Retain-EG keeps EG's classifiers fixed and adjusts their mixture weights to minimize overall repair-set withdrawals. On both repair and validation data, each marginal or intersectional TPR gap is capped at the larger of two percentage points and EG's existing gap; overall accuracy may fall by at most half a point. We also bound the $\ell_1$ change in weights by 0.25. The original weights remain feasible, so optimized repair-set withdrawals cannot increase. These constraints limit the intervention but do not guarantee test-set fairness or accuracy. The full linear program and stability bound appear in Appendix **??**.

We compare Retain-EG with repair-only, validation-only, trust-region-only, and exact-match variants using the same saved classifiers. Exact-match preserves each audited repair intersection's TPR and overall repair accuracy. Additional experiments vary constraint budgets and the sample sizes used to estimate mixture weights. These comparisons test how much withdrawal reduction is possible and what fairness or accuracy allowance it requires; the audit continues to treat accuracy and withdrawals as distinct costs.

## 5 Results

### 5.1 Baseline audit

Figure 1 compares marginal and intersectional TPR gaps, accuracy, and withdrawals on repair and test data. Credit begins with a mean test intersectional gap of 2.49 percentage points. The three baselines reduce it to 1.67–2.04 points, with overall accuracy between 80.86% and 81.47% and withdrawals below 1%. COMPAS begins with a substantially larger gap of 26.08 points. Hardt, RTDP, and EG reduce it to 21.17, 16.91, and 20.46 points, respectively. These results make COMPAS the more informative setting for examining the costs of larger fairness interventions.

The reductions in COMPAS disparities have different accuracy costs. Original test accuracy is 65.32%; Hardt, RTDP, and EG achieve 63.44%, 59.80%, and 63.24%, corresponding to drops of 1.88, 5.52, and 2.08 percentage points. RTDP has

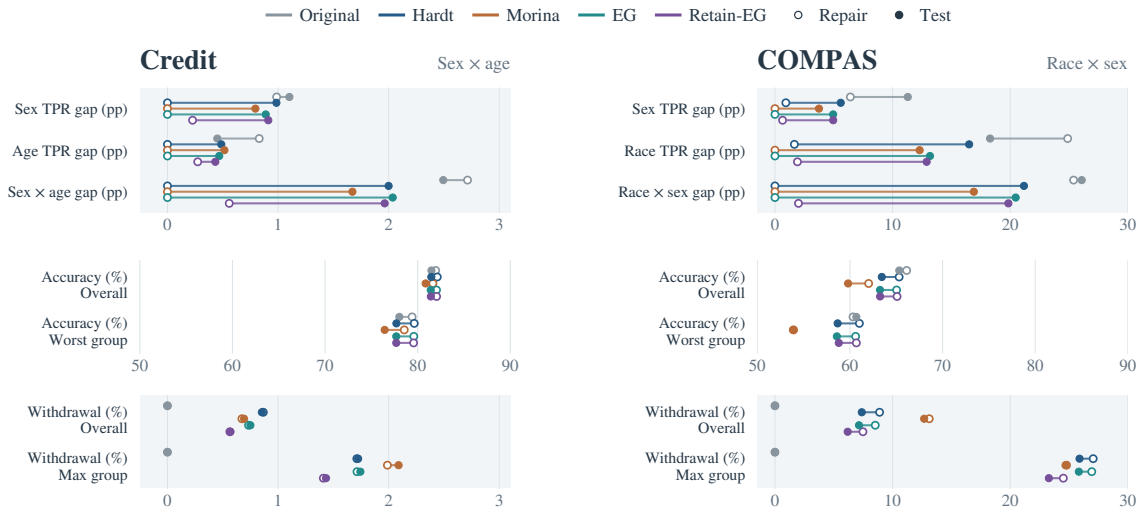



Fig. 1. Fairness, accuracy, and withdrawals on Credit Default and COMPAS. Open and filled points show repair-set and test-set means over two models and five splits. Lines connect the two partitions. Accuracy is absolute; withdrawal is measured relative to original favorable predictions. Worst-group accuracy and maximum-group withdrawal are computed within each fit before averaging. Each metric has its own axis. Retain-EG permits explicit slack in fitted fairness and accuracy.

the smallest test intersectional gap, but also the largest accuracy loss. Worst-group accuracy also declines: from 60.69% originally to 58.67%, 53.85%, and 58.59%, respectively. These minima summarize final performance; they do not identify the largest within-group accuracy drop.

Withdrawal costs also vary and are concentrated within particular groups. Overall COMPAS withdrawal rates are 7.38% for Hardt, 12.69% for RTDP, and 7.14% for EG. The corresponding maximum-group rates are 25.89%, 24.70%, and 25.84%. Thus the baseline with the largest overall withdrawal rate does not have the largest maximum-group rate. Overall averages alone do not show how often originally favorable decisions are reversed for the most affected groups, and the maximizing group can differ across fits.

Comparing repair and test outcomes reveals a further limitation. All three baselines have approximately zero repair-set intersectional gaps, but their COMPAS test gaps remain 16.9–21.2 points. Exact fitted parity therefore transfers imperfectly to new records. Small groups may contribute to this instability: Other men have only 22–31 positive-label repair records across splits. The experiment does not isolate this explanation from variation in scores or policy selection. Marginal and intersectional gaps remain separate audit outcomes, since they describe different group comparisons. Accuracy and withdrawals are likewise evaluated on both partitions; they must be checked directly rather than inferred from a method's fitted fairness constraint.

### 5.2 Retain-EG

Table 1 summarizes the changes from EG to Retain-EG. Mean test withdrawals fall from 0.750% to 0.569% on Credit and from 7.144% to 6.181% on COMPAS. Every paired fit has fewer withdrawals. Average accuracy changes by approximately +0.01 percentage points on each dataset, while mean intersectional gaps decrease by 0.07 and 0.62 points. These averages show that the saved EG rules allow a different balance of outcomes without retraining the underlying score models.

Table 1. Test outcomes of Retain-EG relative to EG, averaged over two models and five splits per dataset. Gaps and accuracy changes are in percentage points (pp); withdrawals are percentages of original favorable predictions. The final column counts fits with no larger intersectional gap, no lower accuracy, and no greater withdrawal.

| Dataset | EG withdrawal | Retain-EG withdrawal | Δ gap (pp) | Δ accuracy (pp) | All three improve or tie |
|---|---|---|---|---|---|
| Credit Default | 0.750% | 0.569% | −0.07 | ≈ +0.01 | 5/10 |
| COMPAS | 7.144% | 6.181% | −0.62 | ≈ +0.01 | 5/10 |

Test gaps nevertheless increase in five Credit fits and two COMPAS fits, and accuracy decreases in four fits per dataset. Retain-EG improves or matches all three outcomes in only half the fits. The corresponding counts against Hardt and RTDP are five and three out of twenty. These comparisons use different policy classes and fitted budgets, rather than a matched fairness target.

The exact-match control provides a stricter comparison. It preserves fitted intersectional TPRs and overall accuracy, and reduces mean test withdrawal to 0.733% on Credit and 6.848% on COMPAS. Some reduction is therefore possible without changing these fitted quantities. However, the gains are smaller than those of Retain-EG, showing that the full method also benefits from its allowed fairness and accuracy slack. The distinction matters when interpreting fewer withdrawals as the removal of unnecessary decision changes.

### 5.3 Sensitivity and subgroup outcomes

The additional constraints affect the two datasets differently. Repair-only reweighting reduces withdrawals further but raises Credit's mean test gap to 2.33 points, compared with 1.96 for Retain-EG. On COMPAS, repair-only reweighting has a slightly smaller gap, 19.66 versus 19.84 points, but lower accuracy, 62.99% versus 63.25%. Neither validation constraints nor the weight bound consistently improves test fairness. Their value depends on which outcome is being protected and on the dataset.

Sensitivity checks show that withdrawal reductions depend more strongly on the fairness allowance than on the accuracy allowance. Removing the two-point fairness floor eliminates about two-thirds of the default mean withdrawal reduction. In contrast, requiring no fitted accuracy decline on either repair or validation data retains most of the benefit. Changing the amount of data used to estimate mixture weights also has mixed effects, with no consistent reduction in test gaps as sample size increases. The detailed budget sweeps, constraint diagnostics, and subsampling results are collected in Appendix **??**.

Subgroup results show why nearly unchanged overall accuracy is insufficient to establish a benefit for everyone. On COMPAS, Retain-EG reduces mean withdrawals by 1.78 points for Caucasian women and 2.17 points for Other men, but by essentially zero for African-American men. Accuracy for the latter group falls by 0.32 points relative to EG. Across subgroup–fit comparisons, accuracy declines in 15 of 60 COMPAS cases and 11 of 40 Credit cases. In one Credit fit, withdrawals for women aged 40 or older rise by 0.40 points even though overall withdrawals fall.

Fewer withdrawals also preserve some incorrect approvals. Of the net additional original approvals retained by Retain-EG relative to EG, 63.5% are correct on COMPAS and 52.8% on Credit under the benchmark labels. Changes in new approvals further offset these gains and losses. This explains why a visible reduction in withdrawals can coexist with little change in overall accuracy. Appendix **??** provides the model-specific and label-specific breakdowns.

## 6 Discussion

Our audit shows why accuracy loss and withdrawals must both be evaluated under an EO target. Smaller test TPR gaps can accompany larger costs, and even identical final fairness and accuracy can involve different withdrawals. Retain-EG demonstrates that some of these decision reversals can be avoided.

Retain-EG addresses one part of this question by finding alternative weights over existing EG rules. It reduces withdrawals across the inspected fits while keeping average accuracy nearly unchanged. The exact-match comparison supports a narrower conclusion: some withdrawal reduction is possible while preserving fitted rates and overall accuracy, but the larger gains of the full method partly depend on allowing fairness slack. Whether this is an acceptable exchange depends on the application's fairness requirement. Our results identify the exchange rather than establish a universally appropriate budget.

The gap analysis also concerns how costs are distributed. Overall accuracy and withdrawal rates can improve while a particular group's accuracy falls or its withdrawals increase. Constraints on overall accuracy do not protect each group's accuracy, and minimizing overall withdrawal does not minimize the worst group's rate. A natural extension would place explicit limits on within-group accuracy loss and withdrawals. Such protection would change the feasible set and could reduce the aggregate retention benefit, which should be measured rather than assumed away.

Generalization remains a separate challenge. Validation constraints limit candidate changes on an additional sample, but they do not make the original validation gaps small or guarantee parity on new records. Likewise, the weight bound limits how much predictions change; it is not a population fairness guarantee. The smaller intersections make rate estimation difficult, and excluding undersupported groups leaves some people outside the reported evaluation. Prospective testing should fix the group definitions, methods, and budgets before evaluating a new sample.

Several features limit the strength of the empirical conclusions. The same records recur across splits, and test results were inspected during project development. We therefore report descriptive paired changes rather than treat the twenty fits as independent replications. The baselines use different policy classes, and Retain-EG permits empirical slack, so the comparisons are not proofs of optimality at a common fairness level. Finally, the datasets provide benchmark labels and hypothetical decisions. They support an audit of prediction changes, but cannot establish the causal welfare consequences of those changes.

## 7 Conclusion

Intersectional EO should be evaluated together with its accuracy and withdrawal costs, both overall and within groups. Our audit finds substantial differences among existing methods and shows that fitted parity can leave large test disparities. Retain-EG reduces withdrawals across all twenty inspected fits with nearly unchanged average accuracy, but its benefits depend on the allowed fairness slack and do not extend uniformly to every subgroup. The results support making both costs explicit when comparing fairness repairs and checking whether the observed balance persists on new data.

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